

- Description

This module manages a smart greenhouse CO2 control system, ensuring that:

1. The system continuously monitors the current CO2 concentration in the greenhouse (in ppm).
2. If the CO2 level falls below a predefined minimum threshold, the CO2 generator is activated to raise the concentration.
3. If the CO2 level exceeds a predefined maximum threshold, the ventilation system is activated to lower the concentration.
4. If the CO2 level is within the optimal range, no action is taken.

This system helps optimize plant growth by maintaining the ideal CO2 concentration in the greenhouse.

- Specification

**module** SmartCO2ControlSystem

**var** currentCO2, minCO2, maxCO2: real

**var** generatorActive, ventilationActive: boolean

**process** updateCO2(newCO2: real) returns (currentCO2: real, generatorActive: boolean, ventilationActive: boolean)

**functional scenario form:**

currentCO2 < minCO2 && generatorActive = true, ventilationActive = false,  
output "ALERT: CO2 level is too low (" + currentCO2 + " ppm). Activating CO2 generator.",  
CO2 generator activated.

||

currentCO2 > maxCO2 && generatorActive = false, ventilationActive = true,  
output "ALERT: CO2 level is too high (" + currentCO2 + " ppm). Activating ventilation  
system.", Ventilation system activated.

||

currentCO2 <= maxCO2 && generatorActive = false, ventilationActive = false,  
output "CO2 level is optimal (" + currentCO2 + " ppm). No action required.", No control  
action required.

**end process**

**end module**

- Java Code

```
package code;
public class c91_SmartCO2Control {
    private double minCO2;
    private double maxCO2;
    private double currentCO2;
    private boolean generatorActive;
    private boolean ventilationActive;
    public c91_SmartCO2Control(double minCO2, double maxCO2, double initialCO2) {
```

```

        this.minCO2 = minCO2;
        this.maxCO2 = maxCO2;
        this.currentCO2 = initialCO2;
        this.generatorActive = false;
        this.ventilationActive = false;
    }
    public void updateCO2(double newCO2) {
        currentCO2 = newCO2;
        adjustControlSystem();
        System.out.printf("Current CO2 Concentration: %.2f ppm\n", currentCO2);
        if (generatorActive) {
            System.out.println("ALERT: CO2 level too low. Activating CO2 generator.");
        } else if (ventilationActive) {
            System.out.println("ALERT: CO2 level too high. Activating ventilation system.");
        } else {
            System.out.println("CO2 level is optimal. No action required.");
        }
    }
    private void adjustControlSystem() {
        if (currentCO2 < minCO2) {
            generatorActive = true;
            ventilationActive = false;
        } else if (currentCO2 > maxCO2) {
            generatorActive = false;
            ventilationActive = true;
        } else {
            generatorActive = false;
            ventilationActive = false;
        }
    }
    public double getCurrentCO2() {
        return currentCO2;
    }

    public boolean isGeneratorActive() {
        return generatorActive;
    }

    public boolean isVentilationActive() {
        return ventilationActive;
    }

```

```

    }

    public static void main(String[] args) {
        c91_SmartCO2Control co2Control = new c91_SmartCO2Control(350.0, 80
0.0, 500.0);

        // testcase-VT:
        //     co2Control.updateCO2(300.0);
        //     co2Control.updateCO2(750.0);
        //     co2Control.updateCO2(850.0);

        // testcase-FT:
        //     co2Control.updateCO2(1.7976931348623157E308);
        //     co2Control.updateCO2(0.0);
        //     co2Control.updateCO2(9.745314011399998E288);

        // testcase-Z3:
        //     co2Control.updateCO2(300.0);
        //     co2Control.updateCO2(750.0);
        //     co2Control.updateCO2(850.0);

        // testcase-UVT:
        co2Control.updateCO2(300.0);
        co2Control.updateCO2(750.0);
        co2Control.updateCO2(850.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

| Element                       | Coverage | Covered Branches | Missed Branches | Total Branches |
|-------------------------------|----------|------------------|-----------------|----------------|
| ✓ VibrationTestingTool        | 1.0 %    | 8                | 834             | 842            |
| ✓ src                         | 1.0 %    | 8                | 834             | 842            |
| ✓ code                        | 1.0 %    | 8                | 834             | 842            |
| ✓ c91_SmartCO2Control.java    | 100.0 %  | 8                | 0               | 8              |
| ✓ c91_SmartCO2Control         | 100.0 %  | 8                | 0               | 8              |
| main(String[])                | 100.0 %  | 0                | 0               | 0              |
| c91_SmartCO2Control(double, c | 100.0 %  | 0                | 0               | 0              |
| adjustControlSystem()         | 100.0 %  | 4                | 0               | 4              |
| updateCO2(double)             | 100.0 %  | 4                | 0               | 4              |

3. Vibration testing instruction coverage: 92%

| Element                       | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ src                         | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ code                        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ c91_SmartCO2Control.java    | 92.0 %   | 103                  | 9                   | 112                |
| ▼ c91_SmartCO2Control         | 92.0 %   | 103                  | 9                   | 112                |
| main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |
| c91_SmartCO2Control(double, c | 100.0 %  | 18                   | 0                   | 18                 |
| adjustControlSystem()         | 100.0 %  | 33                   | 0                   | 33                 |
| updateCO2(double)             | 100.0 %  | 35                   | 0                   | 35                 |

#### 4. Fuzz testing branch coverage: 75%

| Element                       | Coverage | Covered Branches | Missed Branches | Total Branches |
|-------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool        | 0.7 %    | 6                | 836             | 842            |
| ▼ src                         | 0.7 %    | 6                | 836             | 842            |
| ▼ code                        | 0.7 %    | 6                | 836             | 842            |
| ▼ c91_SmartCO2Control.java    | 75.0 %   | 6                | 2               | 8              |
| ▼ c91_SmartCO2Control         | 75.0 %   | 6                | 2               | 8              |
| main(String[])                |          | 0                | 0               | 0              |
| c91_SmartCO2Control(double, c |          | 0                | 0               | 0              |
| adjustControlSystem()         | 75.0 %   | 3                | 1               | 4              |
| updateCO2(double)             | 75.0 %   | 3                | 1               | 4              |

#### 5. Fuzz testing instruction coverage: 83.9%

| Element                       | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool        | 0.4 %    | 94                   | 21,541              | 21,635             |
| ▼ src                         | 0.4 %    | 94                   | 21,541              | 21,635             |
| ▼ code                        | 0.4 %    | 94                   | 21,541              | 21,635             |
| ▼ c91_SmartCO2Control.java    | 83.9 %   | 94                   | 18                  | 112                |
| ▼ c91_SmartCO2Control         | 83.9 %   | 94                   | 18                  | 112                |
| main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |
| c91_SmartCO2Control(double, c | 100.0 %  | 18                   | 0                   | 18                 |
| adjustControlSystem()         | 81.8 %   | 27                   | 6                   | 33                 |
| updateCO2(double)             | 91.4 %   | 32                   | 3                   | 35                 |

#### 6. Z3 branch coverage: 100%

| Element                       | Coverage | Covered Branches | Missed Branches | Total Branches |
|-------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool        | 1.0 %    | 8                | 834             | 842            |
| ▼ src                         | 1.0 %    | 8                | 834             | 842            |
| ▼ code                        | 1.0 %    | 8                | 834             | 842            |
| ▼ c91_SmartCO2Control.java    | 100.0 %  | 8                | 0               | 8              |
| ▼ c91_SmartCO2Control         | 100.0 %  | 8                | 0               | 8              |
| main(String[])                |          | 0                | 0               | 0              |
| c91_SmartCO2Control(double, c |          | 0                | 0               | 0              |
| adjustControlSystem()         | 100.0 %  | 4                | 0               | 4              |
| updateCO2(double)             | 100.0 %  | 4                | 0               | 4              |

#### 7. Z3 instruction coverage: 92%

| Element                       | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ src                         | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ code                        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ c91_SmartCO2Control.java    | 92.0 %   | 103                  | 9                   | 112                |
| ▼ c91_SmartCO2Control         | 92.0 %   | 103                  | 9                   | 112                |
| main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |
| c91_SmartCO2Control(double, c | 100.0 %  | 18                   | 0                   | 18                 |
| adjustControlSystem()         | 100.0 %  | 33                   | 0                   | 33                 |
| updateCO2(double)             | 100.0 %  | 35                   | 0                   | 35                 |

#### 8. Unimproved V-method branch coverage: 100%

| Element                       | Coverage | Covered Branches | Missed Branches | Total Branches |
|-------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool        | 1.0 %    | 8                | 834             | 842            |
| ▼ src                         | 1.0 %    | 8                | 834             | 842            |
| ▼ code                        | 1.0 %    | 8                | 834             | 842            |
| ▼ c91_SmartCO2Control.java    | 100.0 %  | 8                | 0               | 8              |
| ▼ c91_SmartCO2Control         | 100.0 %  | 8                | 0               | 8              |
| main(String[])                |          | 0                | 0               | 0              |
| c91_SmartCO2Control(double, c |          | 0                | 0               | 0              |
| adjustControlSystem()         | 100.0 %  | 4                | 0               | 4              |
| updateCO2(double)             | 100.0 %  | 4                | 0               | 4              |

## 9. Unimproved V-method instruction coverage: 92%

| Element                       | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ src                         | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ code                        | 0.5 %    | 103                  | 21,532              | 21,635             |
| ▼ c91_SmartCO2Control.java    | 92.0 %   | 103                  | 9                   | 112                |
| ▼ c91_SmartCO2Control         | 92.0 %   | 103                  | 9                   | 112                |
| main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |
| c91_SmartCO2Control(double, c | 100.0 %  | 18                   | 0                   | 18                 |
| adjustControlSystem()         | 100.0 %  | 33                   | 0                   | 33                 |
| updateCO2(double)             | 100.0 %  | 35                   | 0                   | 35                 |